

ESP8266 6-Channel Relay Controller

Comprehensive Firmware Analysis & Documentation

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Document Overview

This document provides exhaustive analysis of the ESP8266 6-channel relay controller firmware. It covers all functions, data structures, state machines, configuration systems, timing constants, API endpoints, and reliability features implemented for multi-year unattended operation with self-healing capabilities and extensive monitoring.

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1. System Overview & Architecture

Platform: ESP8266 WiFi-enabled microcontroller (32-bit)
Primary Function: Smart relay controller with 6 channels (expandable to 16)
Features: WiFi connectivity, NTP time sync, DS3231 RTC backup, relay scheduling, web UI, REST API, self-healing recovery, factory reset capability
Deployment Model: Long-term unattended operation (7+ years) on UPS-backed hardware
Network Architecture: Dual-mode WiFi (STA + AP simultaneously)
Persistence: LittleFS flash storage with atomic writes
Safety: Watchdog timers, heap monitoring, connection limits, rate limiting

2. Timing Constants & Intervals (milliseconds)

Constant	Value (ms)	Purpose
NTP_RETRY_INTERVAL	30,000	Retry NTP synchronization
WIFI_CHECK_INTERVAL	5,000	Check WiFi connection status
WIFI_CONNECT_TIMEOUT	15,000	WiFi connection timeout
RTC_UPDATE_INTERVAL	100	Update internal RTC
CONFIG_SAVE_MIN_INTERVAL	30,000	Minimum config save interval
CONFIG_SAVE_INTERVAL	300,000	Normal config save interval (5 min)
MAX_DRIFT_CALIBRATION_INTERVAL	2,592,000,000	Max drift calibration (30 days)
PROACTIVE_MAINTENANCE_INTERVAL	60,000	Maintenance check (1 min)
GENTLE_NETWORK_HEAL_INTERVAL	30,000	Network healing interval
HEAP_DEFRAG_INTERVAL	1,800,000	Heap defragmentation (30 min)
CONNECTION_CLEANUP_INTERVAL	300,000	Clean idle connections (5 min)
IDLE_CONNECTION_TIMEOUT	60,000	Connection idle timeout (1 min)
DS3231_SYNC_INTERVAL	3,600,000	DS3231 sync interval (1 hour)
DEEP_HEAL_INTERVAL	86,400,000	Deep healing (24 hours)
FLASH_WEAR_CHECK_INTERVAL	604,800,000	Flash wear check (7 days)
CRYSTAL_CALIBRATION_MIN_INTERVAL	3,600,000	Crystal calibration (1 hour)

3. Stability & Safety Thresholds

Parameter	Value	Purpose
MIN_HEAP_FOR_SAVE	5,120 bytes	Minimum heap before config save
MIN_HEAP_FOR_NTP	8,192 bytes	Minimum heap for NTP sync
MAX_RECONNECT_ATTEMPTS	20	Max WiFi reconnection attempts
RECONNECT_COOLDOWN	3,600,000 ms	WiFi reconnect cooldown (1 hour)
HEAP_CHECK_INTERVAL	1,800,000 ms	Heap check frequency (30 min)
SELF_CHECK_INTERVAL	86,400,000 ms	Self-check frequency (24 hours)
LOOP_WATCHDOG_TIMEOUT	300,000 ms	Main loop watchdog (5 min)
MAX_SIMULTANEOUS_CONNECTIONS	4	Max concurrent HTTP connections
API_RATE_LIMIT	10 req/sec	API request rate limit
CRITICAL_HEAP_THRESHOLD	8,192 bytes	Critical heap level
EMERGENCY_HEAP_THRESHOLD	5,120 bytes	Emergency heap level

4. Data Structures & Configuration

Struct: RateLimiter

```
unsigned long windowStart = 0;
uint8_t requestCount = 0;
```

Struct: TimerSchedule

```
uint8_t startHour[8], startMinute[8], startSecond[8];
uint8_t stopHour[8], stopMinute[8], stopSecond[8];
bool enabled[8];
uint8_t days[8];
uint32_t monthDays[8];
```

Struct: RelayConfig

```
TimerSchedule schedule;
bool manualOverride;
bool manualState;
char name[16];
uint8_t pin;
bool activeLow;
```

Struct: SystemConfig

```
uint16_t magic;
uint8_t version;
char sta_ssid[32];
char sta_password[64];
char ap_ssid[32];
char ap_password[32];
char ntp_server[48];
long gmt_offset;
int daylight_offset;
```

Struct: ExtConfig

```
uint8_t magic;
uint8_t ap_channel;
uint8_t ntp_sync_hours;
uint8_t ap_hidden;
uint8_t reserved[28];
```

Struct: PinConfig

```
uint8_t magic;
uint8_t numRelays;
bool globalActiveLow;
uint8_t reserved[29];
```

5. Enums & State Machines

Enum: FactoryResetState

FR_IDLE, FR_INIT_LED, FR_DELETE_FILES, FR_CLEAR_CONFIG,
FR_RESET_WIFI, FR_SETUP_AP, FR_DNS_RESTART, FR_WEB_RESTART, FR_COMPLETE

Enum: NTPState

NTP_IDLE, NTP_REQUESTING, NTP_PROCESSING
--

Enum: WiFiHealState

WH_IDLE, WH_DISCONNECT, WH_CONNECTING, WH_WAITING

Enum: SelfRecoveryState

SR_IDLE,
SR_MEMORY_CLEANUP,
SR_WIFI_RECONNECT,
SR_WEB_SERVER_RECOVER,
SR_DNS_RECOVER,
SR_RTC_RECOVER,
SR_TIMER_RECOVER,

Enum: WifiConnState

WCS_IDLE, WCS_PENDING

Enum: APRestartState

APR_IDLE, APR_PENDING, APR_RESTARTING

6. Complete Function Reference

Total Functions Documented: 165

Time Synchronization

getCurrentEpoch, syncInternalRTC, syncInternalRTCFromBrowser, updateCrystalCalibration, syncFromDS3231, syncToDS3231

Configuration Management

loadConfiguration, saveConfiguration, loadExtConfig, saveExtConfig, loadPinConfig, savePinConfig, markConfigDirty, flushConfigIfNeeded

WiFi Management

beginWiFiConnect, startMDNS, enableWiFiStation, disableWiFiStation, toggleWiFiStation, checkWiFiStability

Self-Recovery

selfRecoveryWifiReconnect, selfRecoveryWebServer, selfRecoveryDNS, selfRecoveryRTC, selfRecoveryMemoryCleanup, selfRecoveryNetworkHeal, selfRecoveryDeepHeal, selfRecoveryFlashMaintenance

NTP & Time

tryNTPSync, processNTPState, initDS3231

Memory & Heap

performHeapCleanup, defragmentHeap, emergencyHeapRecovery, monitorHeapHealth, performSelfCheck

Relay Control

updateRelayPins, processRelaySchedules, preserveRelayState, restoreRelayState

Maintenance

performDeepHeal, performFlashMaintenance, proactiveMaintenance, gentleNetworkHealing, processNetworkHealing

Factory Reset

checkFactoryResetButton, processFactoryReset, performFactoryReset

Web Server

setupWebServer, restartAP, processAPRestart

Utilities

updateStatusLED, manageConnections, cleanupIdleConnections, extractJsonInt, extractJsonBool, loadFromFile, saveToFileAtomic

7. Web API Endpoints

Total Handlers: 39

Endpoint Handler	Purpose
handleApPage	API endpoint
handleBrowserTimeSync	Sync time from browser
handleCaptivePortal	API endpoint
handleConnectTest	API endpoint
handleFactoryReset	Factory reset
handleGetAP	Get AP settings
handleGetNTP	Get NTP settings
handleGetPins	Get GPIO pin config
handleGetRelays	Get relay states
handleGetSystem	Get system info
handleGetTime	Get current time
handleGetWiFi	Get WiFi settings
handleManualControl	Manual relay override
handleNcsi	API endpoint
handleNotFound	API endpoint
handleNtpPage	API endpoint
handlePinsPage	API endpoint
handleRedirect	API endpoint
handleReset	Soft reset device
handleResetManual	Reset manual override
handleResetPins	Reset to default pins
handleRoot	Web UI dashboard
handleSaveAP	Save AP settings
handleSaveNTP	Save NTP settings
handleSavePins	Save GPIO pin config
handleSaveRelay	Configure relay
handleSaveRelayName	API endpoint
handleSaveSystem	Save system config
handleSaveWiFi	Save WiFi credentials
handleSoftReset	Trigger self-recovery

8. Self-Recovery State Machine

The firmware implements a non-blocking state machine for automatic recovery from various failure conditions:

Recovery States:

1. SR_IDLE
2. SR_MEMORY_CLEANUP
3. SR_WIFI_RECONNECT
4. SR_WEB_SERVER_RECOVER
5. SR_DNS_RECOVER
6. SR_RTC_RECOVER
7. SR_TIMER_RECOVER
8. SR_NETWORK_HEAL
9. SR_DEEP_HEAL
10. SR_FLASH_MAINTENANCE

Recovery Flow:

The system monitors various health metrics and transitions through recovery states:

- **Memory Cleanup:** Frees fragmented heap and resets JSON buffers
- **WiFi Reconnect:** Attempts WiFi station reconnection with exponential backoff
- **Web Server Recovery:** Restarts HTTP/DNS servers if they fail
- **RTC Recovery:** Reinitializes DS3231 if communication fails
- **Network Healing:** Performs gentle WiFi/DNS recovery without full reset
- **Deep Healing:** Comprehensive system check and reset (24-hour interval)
- **Flash Maintenance:** Validates flash integrity and manages wear leveling

9. Global State Variables

Section:

```
unsigned long factoryResetPressStart = 0;
bool factoryResetButtonPressed = false;
bool factoryResetInProgress = false;
uint8_t factoryResetLEDStep = 0;
unsigned long factoryResetLEDTimer = 0;
unsigned long frStepTimer = 0;
unsigned long ntpTimer = 0;
unsigned long whTimer = 0;
```

10. Critical Features & Safety Systems

1. Multi-level Time Synchronization:

- NTP primary with 4-server fallback (Google, Windows, Cloudflare, Facebook)
- DS3231 RTC backup with battery power
- Browser time sync as secondary fallback
- Crystal drift calibration (PPM correction)
- Monotonic time with millis() overflow tracking (49.7-day rollover handling)

2. Dual-Mode WiFi:

- STA mode: Connects to external network (configurable)
- AP mode: Always active access point for local configuration
- Simultaneous operation (WIFI_AP_STA mode)
- mDNS advertising (hostname.local)
- 30-second AP liveness watchdog

3. Smart Relay Scheduling:

- 8 independent schedules per relay channel
- Time-based triggers (hour, minute, second)
- Day-of-week bitmask support (Sunday-Saturday)
- Month-day support (32-bit mask)
- Manual override capability
- Relay state persistence across reboots

4. Memory Protection:

- Heap fragmentation monitoring
- Automatic cleanup when free heap < 5,120 bytes
- Emergency response at critical levels
- Response simplification under low-memory conditions
- Deferred configuration saves

5. Flash Wear Reduction:

- Dirty-flag batching (300× write reduction)
- Atomic writes with temporary backup files
- Write rate limiting (max 100/hour)
- Weekly flash integrity checks

6. Rate Limiting & Protection:

- API: 10 requests/second per window
- Page requests: 20 requests/second
- Max 4 simultaneous HTTP connections
- 60-second idle connection timeout
- WiFi reconnection backoff (max 20 attempts, 1-hour cooldown)

7. Watchdog & Timeout Mechanisms:

- Main loop watchdog (5-minute timeout)
- WiFi connection timeout (15 seconds)
- NTP retry interval (30 seconds)
- Connection cleanup (5-minute interval)

11. Configuration & Persistence

Configuration Files (LittleFS):

/system.cfg (SystemConfig struct)

- WiFi credentials (SSID/password)
- AP settings
- NTP server configuration
- GMT/daylight offsets
- RTC epoch backup
- Device hostname
- Crystal calibration (PPM)

/ext.cfg (ExtConfig struct)

- WiFi AP channel selection
- NTP sync interval (hours)
- AP visibility (hidden/open)

/relays.cfg (RelayConfig array)

- Per-relay timer schedules (8 per relay)
- Manual override flags
- Custom relay names

/pins.cfg (PinConfig struct)

- GPIO pin assignments (up to 16 relays)
- Active-low/active-high configuration

/state.bak (RelayStateBackup struct)

- Current relay states (for persistence)
- Manual override states

Write Safety:

- All writes use atomic operations
- Temporary backup written first
- Configuration marked "dirty" on change
- Batched writes at 5-minute intervals
- Minimum 30-second save interval

12. Factory Reset & Initialization

Trigger Methods:

1. GPIO0 (FLASH) button held for 5 seconds
2. HTTP API endpoint: /api/factory-reset
3. Web interface reset button

Non-blocking State Machine:

FR_IDLE → FR_INIT_LED → FR_DELETE_FILES → FR_CLEAR_CONFIG
→ FR_RESET_WIFI → FR_SETUP_AP → FR_DNS_RESTART → FR_WEB_RESTART → FR_COMPLETE

Actions During Reset:

1. LED blinks fast during process
2. All LittleFS configuration files deleted
3. WiFi credentials cleared
4. WiFi reset to AP mode with defaults
5. DNS and Web servers restarted
6. Device boots with factory defaults

Post-Reset Defaults:

- AP SSID: "ESP8266_6CH_Timer_Switch"
- AP Password: "ESP8266-admin"
- Default GPIO: {D0/16, D5/14, D6/12, D7/13, RX/3, TX/1}
- All relays disabled
- NTP enabled with Google time server

13. Multi-Year Reliability Engineering

Time Precision & Accuracy:

- System supports year 2106 (32-bit epoch extension)
- Handles millis() 49.7-day rollover safely
- Tracks overflow count for monotonic time
- Drift compensation via crystal calibration
- Safe elapsed-time calculations (no absolute deadline patterns)

Memory Efficiency:

- Avoids dynamic allocations in high-frequency handlers
- Stack-allocated buffers instead of Arduino String
- Response simplification under low-memory (400-byte threshold)
- Heap defragmentation every 30 minutes

Flash Wear Management:

- Estimated 7-year continuous operation (with write batching)
- Previous iteration: ~6 months without optimization
- Current: 300× reduction in flash writes via dirty-flag batching
- Rate-limited to 100 writes/hour maximum

Network Stability:

- Non-blocking WiFi recovery state machine
- Exponential backoff on connection failures
- Gentle healing vs. deep healing mechanisms
- AP interface always active (survives STA disconnections)
- 30-second AP liveness watchdog

Monitoring & Diagnostics:

- Real-time heap monitoring
- Connection count tracking
- WiFi signal strength (RSSI) reporting
- System uptime tracking
- NTP sync age tracking
- DS3231 temperature monitoring

Safety Margins:

- Critical heap threshold: 8 KB
- Emergency heap threshold: 5 KB
- Min heap for NTP: 8 KB
- Min heap for saves: 5 KB

14. Security & API Notes

■ ■ Security Considerations (Current Implementation):

1. Authentication:

- All API endpoints are OPEN (no authentication)
- Suitable only for local network deployment
- AP network accessible with shared password
- WiFi credentials stored in plaintext in flash

2. Recommended Deployment Safeguards:

- Deploy on isolated local network only
- Do not expose to WAN/internet
- Use strong WiFi AP password (default: "ESP8266-admin")
- Change default credentials immediately after boot
- Monitor network access via gateway/firewall

3. Input Validation:

- Rate limiting on API endpoints
- JSON deserialization with bounds checking
- String length validation on user inputs
- Hostname limited to 31 characters
- Relay names limited to 15 characters

4. Denial of Service (DoS) Protections:

- API rate limit: 10 requests/second
- Page rate limit: 20 requests/second
- Max 4 simultaneous connections
- 60-second idle connection timeout
- Automatic connection cleanup

15. Diagnostic & Testing Information

Firmware Statistics:

- Total file size: ~3,750 lines of code
- Total functions: 165
- HTTP handlers: 39
- Data structures: 7
- State enums: 6
- Configuration #defines: 14

System Testing Endpoints:

- GET /api/time → Current system time (JSON)
- GET /api/system → System info (heap, uptime, WiFi)
- GET /api/relays → Relay states
- GET /api/wifi → WiFi configuration
- GET /api/ntp → NTP status and configuration
- POST /api/sync-ntp → Force NTP synchronization
- GET /api/reset → Soft reset (triggers self-recovery)

Recommended Monitoring:

- Log /api/system responses hourly for trend analysis
- Monitor free heap (target: >15 KB)
- Verify NTP sync age (should be <1 hour)
- Check WiFi RSSI (stronger = better reliability)
- Validate relay schedule execution

Debug Serial Output:

- Baud rate: 115,200
- Enable serial monitoring during development
- Watch for memory allocation/deallocation patterns